11-03 第二篇

Coral Reefs: Underwater Cities in Danger

Coral reefs are often called the rainforests of the sea. Although they cover less than one percent of the ocean floor, they are home to about 25 percent of all marine species. Fish, sea turtles, sharks, and countless smaller creatures depend on reefs for food, shelter, and breeding grounds. The bright colors and complex structures of coral reefs make them some of the most beautiful ecosystems on the planet. [1] Coral reefs are incredibly rich in biodiversity, and their health is closely linked to the health of the entire ocean ecosystem.

Corals are actually tiny animals called polyps. They build hard skeletons of calcium carbonate, and over hundreds of years, these skeletons pile up to form the massive structures we know as reefs. Corals have a special relationship with tiny algae that live inside them. The algae provide corals with food through photosynthesis, and in return, the corals give the algae a safe place to live. [2] This partnership between corals and algae is the foundation of reef life, and when it breaks down, the consequences for the entire reef can be devastating.

Climate change is the biggest threat to coral reefs. When ocean temperatures rise even slightly, corals become stressed and expel their algae, turning white in a process called bleaching. If the water stays too warm for too long, the corals die. Ocean acidification, caused by the absorption of excess carbon dioxide, also weakens coral skeletons. [3] The combined effects of warming and acidification are destroying reefs at an alarming rate, and scientists warn that without urgent action, most of the world's coral reefs could be lost within decades.

Human activities also harm reefs directly. Pollution from farms and cities washes into the sea, smothering corals with sediment and nutrients that cause harmful algal blooms. Overfishing removes species that keep reefs in balance. Careless tourism—anchoring on reefs, touching corals, or using chemical sunscreen—adds to the damage. [4] Reducing these local pressures is essential because healthy reefs are better able to withstand the global threats of climate change and ocean acidification.

Efforts to protect coral reefs are growing. Marine protected areas limit fishing and development near sensitive reefs. Scientists are breeding heat-resistant corals and transplanting them to damaged areas. Communities are learning to manage their reefs sustainably. [5] While the challenges are enormous, the combination of scientific innovation, policy action, and community involvement gives hope that coral reefs can be preserved for future generations to study, enjoy, and rely upon.

选项

A. Coral reefs are incredibly rich in biodiversity, and their health is closely linked to the health of the entire ocean ecosystem. B. This partnership between corals and algae is the foundation of reef life, and when it breaks down, the consequences for the entire reef can be devastating. C. The combined effects of warming and acidification are destroying reefs at an alarming rate, and scientists warn that without urgent action, most of the world's coral reefs could be lost within decades. D. Reducing these local pressures is essential because healthy reefs are better able to withstand the global threats of climate change and ocean acidification. E. While the challenges are enormous, the combination of scientific innovation, policy action, and community involvement gives hope that coral reefs can be preserved for future generations to study, enjoy, and rely upon. F. For instance, Australia's Great Barrier Reef, the largest coral reef system in the world, experienced severe bleaching events in 2016, 2017, and 2020, affecting more than half of its corals.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前说珊瑚礁被称为海洋雨林——覆盖不到1%海底但拥有25%海洋物种、鱼龟鲨等依赖珊瑚礁、色彩缤纷结构复杂，空后需概括珊瑚礁的生态价值。A的"incredibly rich in biodiversity"与"25 percent of all marine species"呼应，"health is closely linked to the health of the entire ocean ecosystem"升华生态意义。
- **[2] B**：
  - **线索**：空前说珊瑚是小动物珊瑚虫、建造碳酸钙骨架形成礁石、与体内藻类共生——藻类光合提供食物、珊瑚提供栖所，空后需总结共生关系的重要性。B的"This partnership between corals and algae is the foundation of reef life"概括前文共生关系，"when it breaks down, the consequences... can be devastating"与下文白化现象自然衔接。
- **[3] C**：
  - **线索**：空前说气候变化是最大威胁——海温升高导致白化、长期高温导致死亡、海洋酸化削弱骨架，空后需总结这些威胁的严重性。C的"combined effects of warming and acidification"概括前文两大威胁，"destroying reefs at an alarming rate"与"most... could be lost within decades"给出紧迫警告。
- **[4] D**：

- **线索**：空前说人类活动直接伤害——农业城市污染、过度捕捞、不当旅游，空后需总结减少本地压力的必要性。D的"Reducing these local pressures is essential"针对前文具体人为伤害，"healthy reefs are better able to withstand the global threats"将本地行动与全球挑战联系起来。

- **[5] E**：
  - **线索**：空前说保护努力增长——海洋保护区、培育耐热珊瑚移植、社区可持续管理，空后需总结保护前景。E的"combination of scientific innovation, policy action, and community involvement"概括前文三方面努力，"gives hope that coral reefs can be preserved for future generations"是全文收尾的积极展望。

- **干扰项分析**：
  - **F. For instance, Australia's Great Barrier Reef, the largest coral reef system in the world, experienced severe bleaching events in 2016, 2017, and 2020, affecting more than half of its corals.**：以For instance开头，为具体珊瑚礁案例（大堡礁白化事件）。各空均为概括或过渡句；[3]易被误选因涉及白化，但该空需总结变暖和酸化的整体破坏性，而非大堡礁的具体白化数据。